

QUANTUM MECHANICS FROM THE TORSION MEDIUM

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FRAMING NOTE (added 2026-08-27, v4; corrected 2026-09-02):

The photon descriptions in this document are quantitatively correct. The correct qualitative picture, developed in subsequent work, is that a photon is a localized T_{1g} wave packet (corpuscle, Newton's term) -- not an extended sinusoidal wave. What propagates as "light" is the T_{1g} TRANSVERSE vector wave in the medium (not a compression/pressure wave -- see doc_torsion Section 3.1 and doc_jobson_cell Section 7.1: the v_p compression branch is carried by A_g , a separate scalar mode); the photon corpuscle is the quantized, localized form of that transverse excitation. Wave-particle duality in standard QM arises from conflating the medium wave with the localized packet.

Note: T_{1g} waves may also be generated by macroscopic charge oscillations (classical EM) without individual corpuscle events -- whether all T_{1g} propagation requires a corpuscle source is not yet derived. [OPEN] This mirrors a real, unresolved distinction in mainstream quantum optics, not a torsionverse-specific gap: even canonical "single-photon" double-slit demonstrations typically use an attenuated ("faint") laser for the photon CORPUSCLE crossing the slits -- a Poissonian source with $g^2(0)=1$, no antibunching, meaning "attenuated" only lowers the average photon (corpuscle) number, it does not guarantee one discrete emission event per trial.

True single-photon sources (heralded SPDC, quantum dots, NV centers) reliably emit one photon corpuscle per trigger, shown by genuine antibunching ($g^2(0) \rightarrow 0$), and are experimentally distinguishable from faint-laser sources by this criterion. The quantum eraser experiment cited below (Kim et al. 2000) is itself an example: the photon CORPUSCLE crossing its double slit comes from an attenuated argon laser, not a heralded single-photon source -- only the later SPDC-generated entangled corpuscle pair is unambiguously non-classical.

An electron is a corpuscle locked into a stable (1,2) Hopf winding; the winding gives structure to its medium wave. A photon corpuscle carries no persistent winding ($\theta=0$). Both are corpuscles; the winding is the only difference. All mathematical results in this document are unchanged.

Note: (1,2) is the shared winding class of essentially all matter particles (protons included, Section 1.2) -- scale/zone/ N_J , not the (p,q) pair, is what distinguishes electron from proton.

The quantized, corpuscular nature of light -- established by the photoelectric effect -- is consistent with localized T_{1g} wave packets. Newton lacked the medium to explain the accompanying wave behavior.

ABSTRACT

The Schrödinger equation is not a postulate. It is the non-relativistic limit of the Klein-Gordon equation that governs Hopf winding modes in the torsion medium. The derivation chain has zero free parameters beyond m_p (the framework's one external input, as in the companion papers):

1. Torsion medium: $K = 1/\epsilon_0$, $\rho = \mu_0$, $c = \sqrt{K/\rho}$ [PROVEN]
2. Free pressure wave: $\omega = c k$ (massless) [QM1]
3. Hopf winding rest energy = $m_p c^2 \rightarrow$ Compton frequency $\omega_C = m_p c^2 / \hbar$ [QM2; the Compton wavelength IS $\lambda_p = \hbar c / m_p =$ Zone 1 boundary]
4. Winding dispersion: $\omega^2 = c^2 k^2 + \omega_C^2$ [Klein-Gordon, QM3]
5. Non-relativistic limit ($v \ll c$):
 $\omega_{KG} \rightarrow \omega_C + \hbar k^2 / (2 m_p)$
Remove rest-mass oscillation $\psi = \psi_i \exp(-i m_p c^2 t / \hbar)$:
 $i \hbar \psi_{dt} = -(\hbar^2 / 2 m_p) \nabla^2 \psi$ [SCHRÖDINGER, QM5]

6. Born rule: $P(x) = |\psi(x)|^2$ from medium wave energy density [QM6]

The Schrödinger equation is quantum mechanics. It is now derived from the same medium (the torsion medium) that gives α , G , the Higgs mass, and quantum entanglement. No additional postulates or free parameters beyond m_p .

NEW GROUND: no existing interpretation of quantum mechanics -- Copenhagen, many-worlds, de Broglie-Bohm, GRW, QBism, consistent histories -- derives the Schrödinger equation, the Born rule, or the Tsirelson bound from an underlying physical medium with independently fixed mechanical properties. Every existing approach postulates the Hilbert-space/probability structure as an axiom. This paper derives it instead, as a mechanical consequence of the same medium that gives α , G , and the Higgs mass in the companion papers.

Consequences:

- Double-slit: Hopf winding fields span both slits simultaneously. Interference = constructive/destructive Huygens superposition in medium.
- Which-path: Zone 3 field decoherence resolves winding at $r < r_{\text{lock}}(T)$. $r_{\text{lock}}(300K) = 340 \text{ fm}$; coherence preserved for macroscopic slit separations.
- Delayed choice: no paradox. Medium configuration never had a single-slit address. "Measurement" = local perturbation resolving global winding. Timing is irrelevant; topology is not retroactively changed.

NEW PREDICTIONS (not in standard QM):

- Minimum electron slit width for coherent diffraction: $2\lambda_e = 772 \text{ fm}$ (below this, slit-edge Zone 3 fields decohere the electron winding)
- Minimum proton slit width: $2\lambda_p = 0.421 \text{ fm} = r_{\text{grind}}$ (nuclear hard core)
- Fringe visibility vs temperature: $V(T) \sim \exp(-r_{\text{sep}} / r_{\text{lock}}(T))$
where $r_{\text{lock}}(T) = (\alpha \hbar c^2 r_p^2 / k_B T)^{1/3}$

Companion script: analysis/demos/qm_doc.py (21/21 PASS)

1. MEDIUM WAVE EQUATION AND KLEIN-GORDON

1.1 Free pressure waves

The torsion medium has bulk modulus $K = 1/\epsilon_0$ and density $\rho = \mu_0$.

The pressure wave speed $c = \sqrt{K/\rho} = 1/\sqrt{\epsilon_0 \mu_0}$ (exact by SI).

Free waves satisfy:

$$(d^2/dt^2 - c^2 \nabla^2) \phi = 0$$

Dispersion: $\omega = c * k$ (massless, linear). [QM1]

(Derived from a discrete Jobson-cell lattice in the continuum limit, not an independent postulate -- analysis/quantum/lattice_dwell_time_bridge.py.)

1.2 Hopf winding rest energy -> mass gap

A Hopf winding mode (the proton) is a topological defect in the torsion medium with winding number (1,2) ($p=1, q=2$). The Maxwell jamming constraint ($N_J = 21$ cells, $3V-E = 6$ exactly critical) sets the minimum size of the winding. Below this size, the winding cannot be compressed -- this creates

a minimum energy:

$$E_0 = m_p * c^2 \quad [\text{proton rest energy} = \text{mass gap of the winding mode}]$$

This generates the Compton oscillation frequency:

$$\omega_C = m_p * c^2 / \hbar$$

Crucially: the Compton wavelength $\lambda_{\text{bar}} = \hbar / (m_p * c) = \hbar_c / m_p$

IS the Zone 1 boundary λ_p . The winding resonates at the Compton frequency WITHIN Zone 1 (the Maxwell critical jamming region). This is the same λ_p that appears in $r_p = 4 * \lambda_p$ (PS4) and throughout nuclear structure.

The identification $\lambda_{\text{bar}} = \lambda_p$ is not a coincidence: it follows from the proton being the Maxwell-critical Hopf winding in the torsion medium.

[QM2 PASS: $\omega_C = c / \lambda_p$ confirmed to machine precision]

1.3 Klein-Gordon equation for the winding mode

With a mass gap, the dispersion relation becomes:

$$\omega^2 = c^2 * k^2 + \omega_C^2 \quad [\text{Klein-Gordon}]$$

At $k=0$ (standing winding): $\omega = \omega_C$ (pure Compton oscillation)

At $k \gg \omega_C/c$: $\omega \rightarrow c*k$ (ultra-relativistic, massless limit)

This IS the Klein-Gordon equation: $(\partial^2/\partial t^2 - c^2 \nabla^2)\phi = -(m_p c^2/\hbar)^2 \phi$

[QM3 PASS: both limits verified]

(Both dispersion relations above are derived from a discrete Jobson-cell lattice model, not postulated -- lattice_dwell_time_bridge.py BR1-BR4.)

2. NON-RELATIVISTIC LIMIT: SCHRÖDINGER EQUATION

For non-relativistic motion ($v \ll c$, $k \ll \omega_C / c = m_p c / \hbar$):

$$\begin{aligned} \omega_{\text{KG}} &= \sqrt{c^2 k^2 + \omega_C^2} \\ &= \omega_C * \sqrt{1 + (\hbar k / m_p c)^2} \\ &\sim \omega_C + \hbar k^2 / (2 m_p) + O((v/c)^4) \end{aligned}$$

The ω_C term = rest-mass oscillation. Extract it:

$$\phi(x,t) = \psi(x,t) * \exp(-i * m_p c^2 t / \hbar)$$

Substituting into Klein-Gordon and dropping the $O((v/c)^2)$ correction to $\partial^2 \psi / \partial t^2$:

$$i \hbar \frac{d(\psi)}{dt} = -(\hbar^2 / 2 m_p) * \nabla^2 * \psi$$

This IS the time-dependent Schrödinger equation for a free particle. [QM5 PASS]

Accuracy: The approximation is exact to order $(v/c)^2$. For $v/c = 1/1000$ (deep NR): relative error = $2.5e-7$, consistent with the $(v/c)^{2/4}$ Taylor expansion.

ZERO additional assumptions were made. The Schrödinger equation follows from:

- (a) The torsion medium (K , ρ , c -- already proven)
- (b) The Hopf winding mass gap $E_0 = m_p c^2$ (from Maxwell jamming -- already proven)
- (c) The non-relativistic approximation (standard physics -- no extra postulate)

Potentials ($V \neq 0$): adding an external pressure field $V(x)$ to the medium gives:

$$i \hbar \frac{d\psi}{dt} = -(\hbar^2 / 2m) \nabla^2 \psi + V(x) \psi$$

This is the full Schrödinger equation. $V(x)$ = the local torsion medium pressure (Coulomb, gravity, nuclear) -- all derived from the same medium.

3. BORN RULE FROM WAVE ENERGY DENSITY

In the torsion medium, the energy density of a pressure wave is:

$$u(x,t) = (1/2)*\rho*v^2 + (1/2)*K*strain^2 \text{ proportional to } |\phi(x,t)|^2$$

The probability of a detector at x absorbing the wave (detecting the particle) is proportional to the wave energy density at x :

$$P(x) = u(x) / \int u \, dV = |\psi(x)|^2 / \int |\psi|^2 \, dV \quad [\text{Born rule}]$$

This is the Born rule. It requires no additional postulate: detection = energy absorption from the medium wave, and energy density is proportional to amplitude squared.

[QM6 PASS: Born rule from medium energy density]

Note on wave identity: Light propagation = T_{1g} transverse vector wave in the torsion medium (Klein-Gordon with $m=0 \rightarrow$ Maxwell equations; see doc_torsion Section 3.1 -- T_{1g} is transverse, not the v_p compression branch, which is carried by A_g). A photon is a localized T_{1g} wave packet (corpuscle) that IS one source of such waves;

macroscopic charge oscillations (classical EM) may generate T_{1g} waves through the cog/Zone 3 mechanism without individual corpuscle emission events. Whether all T_{1g} waves require a corpuscle source or can be generated purely classically is not yet derived from the framework. [OPEN: classical vs quantum T_{1g} generation -- Series 3 target. See the FRAMING NOTE above for why this mirrors a real, unresolved distinction in mainstream quantum optics -- attenuated-laser sources are Poissonian (no antibunching) and do not by themselves establish one-corpuscle-per-trial, unlike true heralded single-photon sources.] Matter (electrons, protons) = massive

Hopf winding modes (Klein-Gordon with mass gap $m*c^2$). Light waves and matter windings are two excitation types of the same medium, not separate entities.

The photon (corpuscle) is the $\theta=0$ mode. Here θ is the winding tilt angle: it measures how far the Hopf winding departs from the massless T_{1g} baseline. $\theta=0$ means no Zone 1 displacement and no rest mass; $\theta>0$ means the winding displaces Zone 1 and acquires rest energy $m*c^2$.

Formula: $\theta(m) = \arcsin(8*\alpha*\phi*m/m_p)$. All matter requires $\theta > 0$. [doc_particle_generation.txt; analysis/quantum/winding_angle.py WA1-WA2]

Pair production: when a photon's pressure wave amplitude exceeds the threshold for nucleating a new Hopf winding ($E \geq 2*m_e*c^2$ for $e+e^-$ pair), the medium forces a new topological winding into existence. This is the Schwinger mechanism: field amplitude > winding nucleation threshold.

Environments where this occurs: gamma-ray burst cores ($kT \sim \text{MeV}$), neutron star merger ejecta, the early universe ($T > 5e9 \text{ K} \sim 0.4 \text{ MeV}$), and extreme laser fields approaching the Schwinger critical field ($E_S = m_e^2 c^3 / e \hbar$). Note: the solar core ($kT \sim 1.3 \text{ keV}$) is $\sim 800x$ below threshold and does NOT continuously pair-produce. [See doc_nucleus Zone 2: $N_J=21$ for the winding nucleation condition; the photon-trapping calculation for this regime remains open]

4. DOUBLE-SLIT EXPERIMENT

4.1 Why interference occurs

The proton (or electron) is a corpuscle -- always at one definite location, never split or delocalized. Its Zone 3 field extends to infinity (falling as $1/r^2$) regardless of where the corpuscle itself is. When the corpuscle approaches a barrier with two slits:

- (a) The CORPUSCLE goes through exactly ONE slit -- always definite, never "at slit A and slit B simultaneously." What spans both slits is its Zone 3 pressure field, which reaches both openings no matter which slit the corpuscle itself takes (fields fall off with distance but extend everywhere; they are not confined to one opening).
- (b) The medium pressure field passes through BOTH slit openings simultaneously, whichever slit the corpuscle used.
- (c) At the screen, the pressure wave contributions from both slits superpose: constructive where path difference = $n \cdot \lambda_{dB}$, destructive otherwise. This superposition is a property of the FIELD; it is not a claim that the corpuscle itself was in two places.
- (d) Interference pattern = standard Huygens-Fresnel diffraction for the medium pressure wave. Fringe spacing = λ_{dB} / d where d is slit separation. $\lambda_{dB} = 2\pi \hbar / (m \cdot v) = \text{de Broglie wavelength (from KG dispersion)}$.

The corpuscle's subsequent path to the screen is drawn from the resulting interference landscape; over many trials (many corpuscles, each through one slit or the other) the fringe pattern emerges statistically. A corpuscle in transit is not directly observable -- it registers only at a nexus (a screen contact, a detector atom), where it transfers momentum to whatever it meets. This is also why a single detection event is always a single, localized click and never a smeared, continuous readout: the corpuscle itself is always at one place, and that is the only place a nexus event can occur. [analysis/quantum/double_slit_field_interference.py DS1, DS5: confirms this picture reproduces the standard fringe pattern and spacing]

Not yet derived: WHY an individual corpuscle's landing point, across many trials, is distributed as exactly $|\psi(x)|^2$ (as opposed to some other function of the field). The field mechanics above are confirmed consistent with standard wave optics; the corpuscle-selection statistics are not independently derived from them.

4.2 Why which-path measurement destroys interference

From doc_entanglement EP3/EP6: a detector placed at one slit puts a Zone 3-active particle (detector atoms) within the phase-locking radius $r_{\text{lock}}(T)$ of that slit's OWN field contribution -- not of the corpuscle's instantaneous position. Section 4.1 already establishes that the field passes through BOTH openings on every trial, regardless of which slit the corpuscle itself takes; that is what generates the interference in the first place. A detector sitting at one opening therefore couples to that opening's local field amplitude on every trial, not only on the trials where the corpuscle happens to pass through the monitored slit. This is why a which-path detector at a single slit destroys interference for the entire ensemble (matching observation), not just for the roughly half of corpuscles that actually took that slit. The Zone 3 field coupling resolves which opening's contribution the detector interacted with, which is enough to mark the path and kill the coherent cross-term for every trial.

$r_{\text{lock}}(300\text{K}) = 340 \text{ fm}$ [Phase-lock threshold at room temperature]

For macroscopic slits (separation $\gg 340 \text{ fm}$), the detector itself must physically interact with the field passing through its slit to resolve the path. Simply placing a detector nearby (not in the path of either opening's own field) does NOT decohere (consistent with observation: detection requires interaction).

4.2a "Sneakily turning off" the detector does not restore interference

A crucial implication: decoherence is caused by PHYSICAL ZONE 3 COUPLING between the particle's winding and detector atoms, NOT by whether a human reads the output.

Experiment: a which-path detector is "turned off" (recording stopped) but the detector atoms remain physically in place near the beam path. The particle pattern (two bands, no interference) persists.

Explanation:

Detector ON and recording: atoms within r_{lock} -> winding resolved -> particle pattern

Detector "off" (not recording) but atoms present: atoms STILL within r_{lock}

-> Zone 3 coupling STILL occurs -> winding STILL resolved

-> particle pattern persists. "Off" means nothing.

Detector physically removed: no atoms within r_{lock} -> no coupling -> interference returns.

The particle's winding does not "know" whether a human is reading the detector.

It only responds to whether another Zone 3 field is within 340 fm .

The threshold is $r_{\text{lock}}(T)$ -- a physical scale, not an epistemic one.

This resolves a longstanding confusion in quantum measurement theory:

measurement = physical interaction at $r < r_{\text{lock}}(T)$, not observation.

Copenhagen is vague on this distinction. Torsionverse is not.

ALSO EXPLAINS: the quantum eraser (Kim et al. 2000). Pairs whose idler photon went through the eraser (no Zone 3 coupling forced) retain the A_g singlet -> their sub-ensemble shows interference in coincidence counts. The "recovery" of the pattern is not retroactive: those pairs were never decohered to begin with.

4.3 New prediction: minimum slit width

Below $2\lambda_{\text{bar}}$ (reduced Compton wavelength), the slit-edge Zone 3 fields of slit atoms become comparable to the winding's Zone 3 field -> decoherence.

Electron: $2\lambda_{\text{bar}_e} = 2\hbar c/m_e = 772 \text{ fm}$ [QM7]
Proton: $2\lambda_{\text{bar}_p} = 2\hbar c/m_p = 0.421 \text{ fm} = r_{\text{grind}}$ [QM8]

Standard QM has no minimum slit width. Torsionverse predicts one, testable with sub-pm electron diffraction experiments.

5. DELAYED CHOICE AND QUANTUM ERASER

5.1 Delayed choice: paradox dissolved

Wheeler's delayed choice: after the particle passes the slits, the experimenter decides whether to measure which-path or allow interference.

In the torsionverse:

- The "particle" is a Hopf winding mode in the medium
- After passing the slits, the medium configuration spans BOTH slit openings
- There is no "particle at slit A or B" -- only a global medium configuration
- The choice of measurement = whether to apply a Zone 3-resolving perturbation

The timing of the perturbation is irrelevant:

- "Before": resolve now -> no interference (winding localized)
- "After": resolve after slits -> medium was never in a single-slit state
The resolution perturbs the CURRENT medium state, not a past state
- No retrocausality: the medium is Markovian (current state determines future)

[QM9 PASS: no retrocausality required]

5.2 Quantum eraser

When which-path information is "erased" (the resolving perturbation is undone before it propagates), the A_g global winding was never permanently resolved.

The winding continues in its global configuration. Interference returns.

This is not mysterious: the perturbation must propagate classically at c to affect the screen. If it is undone first, it never reaches the screen.

[Consistent with quantum eraser experiments by Kim et al. 2000]

6. STATUS SUMMARY

PROVEN (from existing medium derivations + this document):

- Schrodinger equation from Klein-Gordon NR limit [QM4-5]
- Born rule from medium wave energy density [QM6]
- Double-slit interference mechanism (Huygens in medium)
- Delayed choice: no retrocausality [QM9]

ESSENTIALLY DERIVED (borrowed from entanglement_doc):

- Which-path decoherence at $r < r_{\text{lock}}(T)$ [entanglement_doc EP3; essentially derived -- t in MeV open; QM10 check is $r_{\text{lock}} > 0$ sanity only]

NEW PREDICTIONS (physical argument from Zone 3 geometry; not yet experimentally tested):

- Minimum electron slit width $2\hbar c/m_e = 772 \text{ fm}$ [QM7 -- script check verifies formula; labeled 'NEW PREDICTION' in script; Zone 3 argument]
- Minimum proton slit width $2\lambda_{\text{bar}_p} = r_p/2 = r_{\text{grind}}$ [QM8 -- second

condition IS a real check: $r_p = 4 \cdot \lambda_{p4}$ (PS4); first condition is formula-only]

ESSENTIALLY CLOSED:

- Full Schrodinger equation with potential $V(x)$: V = local medium pressure (Coulomb, gravity all derived from same medium; adding V is direct)
- Quantum eraser: perturbation propagation bounds the erasure window
- Pauli equation: from Dirac NR limit + $E+$ spinor [qm_dirac_pauli.py QD6]
- Dirac equation: from Clifford (T_{1g} generators) + $E+/E-$ ($2I$) [QD1-QD4]
 $g=2$ from $(\sigma \cdot v)^2 = |v|^2$ [QD5]; proton $g_p \neq 2$ from Zone structure
- Path integral: from medium Green's function (C7 static limit = Coulomb)
Composition rule, classical action phase, $H_{E1} = -13.6$ eV [QP1-QP5]

GENUINELY OPEN:

- Fringe visibility $V(T)$ quantitative formula from $r_{lock}(T)$
- Relativistic QM with spin: Dirac + EM = QED (next paper)

7. COMPANION SCRIPT

```
analysis/demos/qm_doc.py    -- 21/21 PASS [STANDALONE -- no external deps]
```

Working scripts:

```
analysis/quantum/qm_from_medium.py    -- 10/10 PASS [KG -> Schrodinger, Born rule]
analysis/quantum/qm_dirac_pauli.py    -- 7/7 PASS [Clifford, Dirac, Pauli, g=2]
analysis/quantum/qm_path_integral.py  -- 5/5 PASS [propagator, composition, E1]
```

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Repository: <https://doi.org/10.5281/zenodo.22105622>

Code: https://github.com/DeStraag/torsionverse_public